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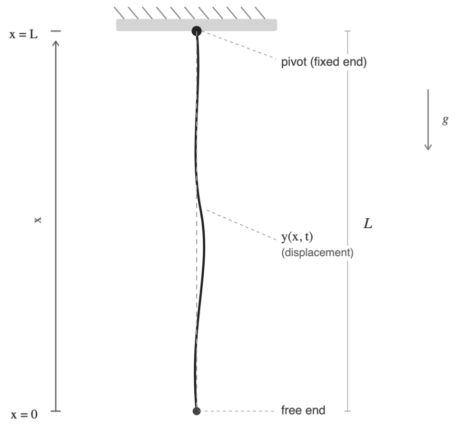
Solution by Pooja

Becoming Best Friends with Bessel



1. Dimensional Analysis

   The given physical parameters are:

   - Mass density: μ $[ML^{-1}]$
   - Length: L $[L]$
   - Gravitational acceleration: g $[LT^{-2}]$

   We seek the dependence of angular frequency ω $[T^{-1}]$.

   Assume:

   $$\omega \propto g^a L^b$$

   Matching dimensions:

   $$[T^{-1}] = (LT^{-2})^a \cdot (L)^b = L^{a+b}T^{-2a}$$

   Equating powers:

   $$-1 = -2a \Rightarrow a = \frac{1}{2}, \quad a + b = 0 \Rightarrow b = -\frac{1}{2}$$

   Thus,

   $$\boxed{\omega \sim \sqrt{\frac{g}{L}}}$$

   This is the same scaling as a simple pendulum.

2. Given differential equation:

   $$x\frac{d^2A}{dx^2} + \frac{dA}{dx} + \frac{\omega^2}{g}A = 0 \tag{1}$$

   Introducing substitution:

   $$u = 2\omega\sqrt{\frac{x}{g}}$$

   $$\frac{du}{dx} = 2\omega \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{gx}} = \frac{\omega}{\sqrt{gx}}$$

Thus,

$$\frac{d}{dx} = \frac{du}{dx}\frac{d}{du} = \frac{\omega}{\sqrt{gx}}\frac{d}{du}$$

$$\frac{d^2}{dx^2} = \frac{d}{dx}\left(\frac{\omega}{\sqrt{gx}}\frac{d}{du}\right)$$

Using product rule:

$$\frac{d^2}{dx^2} = \left(-\frac{\omega}{2\sqrt{g}}x^{-3/2}\right)\frac{d}{du} + \frac{\omega}{\sqrt{gx}}\frac{d}{dx}\left(\frac{d}{du}\right)$$

Now,

$$\frac{d}{dx}\left(\frac{d}{du}\right) = \frac{du}{dx}\frac{d^2}{du^2} = \frac{\omega}{\sqrt{gx}}\frac{d^2}{du^2}$$

Therefore,

$$\frac{d^2}{dx^2} = -\frac{\omega}{2\sqrt{g}}x^{-3/2}\frac{d}{du} + \frac{\omega^2}{gx}\frac{d^2}{du^2}$$

3. Substituting the values into Original Equation:

$$x\frac{d^2A}{dx^2} + \frac{dA}{dx} + \frac{\omega^2}{g}A = 0$$

First term:

$$x\frac{d^2A}{dx^2} = -\frac{\omega}{2\sqrt{g}}x^{-1/2}\frac{dA}{du} + \frac{\omega^2}{g}\frac{d^2A}{du^2}$$

Second term:

$$\frac{dA}{dx} = \frac{\omega}{\sqrt{gx}}\frac{dA}{du}$$

Thus equation becomes:

$$-\frac{\omega}{2\sqrt{g}}x^{-1/2}\frac{dA}{du} + \frac{\omega^2}{g}\frac{d^2A}{du^2} + \frac{\omega}{\sqrt{gx}}\frac{dA}{du} + \frac{\omega^2}{g}A = 0$$

Combining terms:

$$-\frac{\omega}{2\sqrt{g}}x^{-1/2} + \frac{\omega}{\sqrt{gx}} = \frac{\omega}{2\sqrt{gx}}$$

So:

$$\frac{\omega^2}{g}\frac{d^2A}{du^2} + \frac{\omega}{2\sqrt{gx}}\frac{dA}{du} + \frac{\omega^2}{g}A = 0$$

Expressing x in terms of u

From:

$$u = 2\omega\sqrt{\frac{x}{g}} \Rightarrow x = \frac{gu^2}{4\omega^2}$$

Thus:

$$\sqrt{gx} = \frac{gu}{2\omega}$$

Hence:

$$\frac{\omega}{2\sqrt{gx}} = \frac{\omega^2}{gu}$$

Substituting the values and we obtain,

$$\frac{\omega^2}{g}\frac{d^2A}{du^2} + \frac{\omega^2}{gu}\frac{dA}{du} + \frac{\omega^2}{g}A = 0$$

Dividing throughout by $\frac{\omega^2}{g}$:

$$\boxed{\frac{d^2A}{du^2} + \frac{1}{u}\frac{dA}{du} + A = 0}$$

Comparing this eqation with

$$\frac{d^2A}{du^2} + \frac{1}{u}\frac{dA}{du} + \left(1 - \frac{\alpha^2}{u^2}\right) A = 0\,. \tag{2}$$

by substituting $\alpha = 0$, the obtained equation is the Bessel equation of order 0:

$$\frac{d^2A}{du^2} + \frac{1}{u}\frac{dA}{du} + A = 0$$

1 *Why (Circular) Drums Sound "Thuddy" [Substantive]

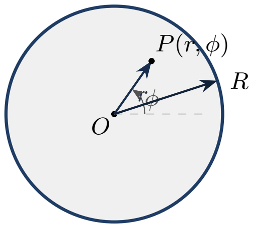


Figure 1: Top-view of the membrane in polar coordinates. The rim at $r = R$ is clamped: $g(R, \phi, t) = 0$.

Given Equation

The 2D wave equation in polar coordinates is:

$$\frac{\partial^2 g}{\partial t^2} = v^2 \left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial g}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 g}{\partial \phi^2}\right], \quad v = \sqrt{\frac{\mathcal{T}}{\sigma}} \tag{3}$$

1. Separation of Variables

(i) Assume:

$$g(r, \phi, t) = R(r)\,\Phi(\phi)\,T(t)$$

Substitute into the wave equation:

$$R\Phi\ddot{T} = v^2\left[\Phi T\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + RT\frac{1}{r^2}\frac{d^2\Phi}{d\phi^2}\right]$$

Divide both sides by $R\Phi T$:

$$\frac{\ddot{T}}{T} = v^2\left[\frac{1}{R}\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \frac{1}{\Phi}\frac{1}{r^2}\frac{d^2\Phi}{d\phi^2}\right]$$

Since, the left side depends only on t, right side only on (r, ϕ), so both must equal a constant:

$$\frac{\ddot{T}}{T} = -\omega^2$$

Thus:

$$\boxed{\ddot{T} + \omega^2 T = 0}$$

Solution:

$$T(t) = A\cos(\omega t) + B\sin(\omega t)$$

(ii) The spatial part can be written as

$$\frac{1}{R}\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \frac{1}{\Phi}\frac{1}{r^2}\frac{d^2\Phi}{d\phi^2} = -\frac{\omega^2}{v^2}$$

Define:

$$k^2 \equiv \frac{\omega^2}{v^2}$$

Multiply by r^2:

$$\frac{r^2}{R}\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = -k^2r^2$$

$$\frac{r^2}{R}\left(R'' + \frac{1}{r}R'\right) + k^2r^2 = -\frac{\Phi''}{\Phi}$$

Since, the left side depends only on r, right side only on ϕ, so both must equal a constant:

$$\frac{r^2}{R}\left(R'' + \frac{R'}{r}\right) + k^2r^2 = -\frac{\Phi''}{\Phi} \equiv m^2, \qquad k \equiv \frac{\omega}{v}.$$

(b) Angular Equation and Quantisation of m

The angular part of the separated equation is:

$$\frac{d^2\Phi}{d\phi^2} + m^2\Phi = 0 \tag{4}$$

(i) Quantisation of m

The general solution of the above equation is:

$$\Phi(\phi) = C\cos(m\phi) + D\sin(m\phi) \tag{5}$$

Since the displacement $g(r, \phi, t)$ must be single-valued, we require:

$$\Phi(\phi + 2\pi) = \Phi(\phi) \tag{6}$$

Substituting $\phi \to \phi + 2\pi$:

$$\Phi(\phi + 2\pi) = C\cos\big(m(\phi + 2\pi)\big) + D\sin\big(m(\phi + 2\pi)\big) \tag{7}$$
$$= C\cos(m\phi + 2\pi m) + D\sin(m\phi + 2\pi m) \tag{8}$$

Using trigonometric identities:

$$\cos(m\phi + 2\pi m) = \cos(m\phi)\cos(2\pi m) - \sin(m\phi)\sin(2\pi m), \tag{9}$$
$$\sin(m\phi + 2\pi m) = \sin(m\phi)\cos(2\pi m) + \cos(m\phi)\sin(2\pi m), \tag{10}$$

we obtain:

$$\Phi(\phi + 2\pi) = C\big[\cos(m\phi)\cos(2\pi m) - \sin(m\phi)\sin(2\pi m)\big] \tag{11}$$
$$+ D\big[\sin(m\phi)\cos(2\pi m) + \cos(m\phi)\sin(2\pi m)\big]. \tag{12}$$

For $\Phi(\phi + 2\pi) = \Phi(\phi)$ to hold for all ϕ, we must have:

$$\cos(2\pi m) = 1, \qquad \sin(2\pi m) = 0. \tag{13}$$

These conditions imply:

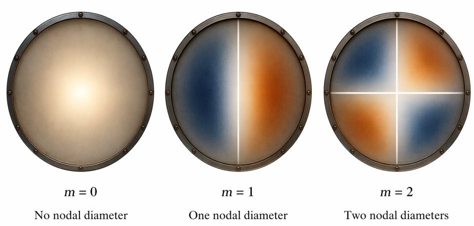
$$m = 0, 1, 2, 3, \ldots \tag{14}$$

Thus, m must be a non-negative integer.

(ii) Nodal Diameters

The integer m determines the angular dependence of the solution and corresponds to the number of nodal diameters (lines through the center where $g = 0$).

- **$m = 0$:** $\Phi =$ constant There is no angular dependence.
  Nodal pattern: No nodal diameters (fully symmetric vibration).
- **$m = 1$:** $\Phi = \cos\phi$ or $\sin\phi$ Zeros occur along one line passing through the center.
  Nodal pattern: One nodal diameter dividing the membrane into two regions.
- **$m = 2$:** $\Phi = \cos(2\phi)$ or $\sin(2\phi)$ Zeros occur along two lines through the center.
  Nodal pattern: Two perpendicular nodal diameters dividing the membrane into four regions.



Nodal patterns for a circular membrane. Bright lines are nodal lines where $g(r, \phi, t) = 0$.

(c) Bessel's Equation

The radial differential equation obtained from separation of variables:

$$r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + (k^2 r^2 - m^2) R = 0. \tag{15}$$

We introduce the substitution:

$$u = kr \quad \Rightarrow \quad r = \frac{u}{k}. \tag{16}$$

Using the chain rule:

$$\frac{d}{dr} = \frac{d}{du}\frac{du}{dr} = k\frac{d}{du}. \tag{17}$$

For the second derivative:

$$\frac{d^2}{dr^2} = \frac{d}{dr}\left(k\frac{d}{du}\right) \tag{18}$$

$$= k\frac{d}{dr}\left(\frac{d}{du}\right) \tag{19}$$

$$= k\left(\frac{d}{du}\frac{d}{du}\right)\frac{du}{dr} \tag{20}$$

$$= k\left(\frac{d^2}{du^2}\right)k \tag{21}$$

$$= k^2\frac{d^2}{du^2}. \tag{22}$$

We substitute each term into the radial equation:

First term:

$$r^2\frac{d^2R}{dr^2} = r^2 \cdot k^2 \frac{d^2R}{du^2}. \tag{23}$$

Since $r = \frac{u}{k}$,

$$r^2 = \frac{u^2}{k^2} \quad \Rightarrow \quad r^2\frac{d^2R}{dr^2} = u^2\frac{d^2R}{du^2}. \tag{24}$$

Second term:

$$r\frac{dR}{dr} = r \cdot k\frac{dR}{du} = \frac{u}{k} \cdot k\frac{dR}{du} = u\frac{dR}{du}. \tag{25}$$

Third term:

$$k^2 r^2 R = k^2 \cdot \frac{u^2}{k^2} R = u^2 R. \tag{26}$$

Substituting all terms into the original equation:

$$u^2\frac{d^2R}{du^2} + u\frac{dR}{du} + (u^2 - m^2)R = 0. \tag{27}$$

Dividing by u^2:

$$\frac{d^2R}{du^2} + \frac{1}{u}\frac{dR}{du} + \left(1 - \frac{m^2}{u^2}\right)R = 0. \tag{28}$$

This is Bessel's equation of order m.
(i) The general solution of Bessel's equation is:

$$R(u) = AJ_m(u) + BY_m(u), \tag{29}$$

where $J_m(u)$ and $Y_m(u)$ are Bessel functions of the first and second kind, which are plotted in Fig 2.

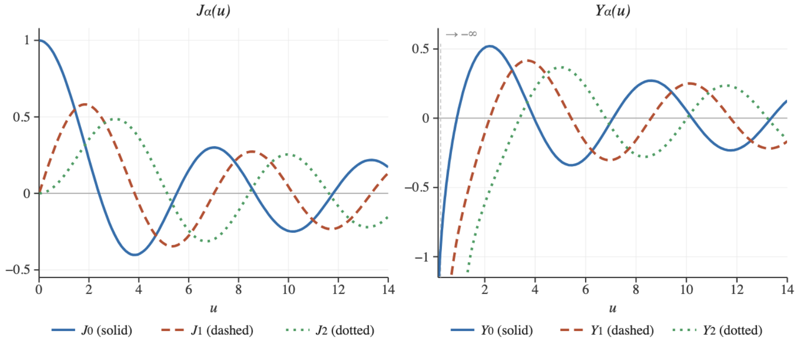


Figure 2: Bessel functions J_m (left) and Y_m (right) for $m = 0, 1, 2$. All J_m are finite at $u = 0$; all Y_m diverge there.

As $u \to 0$ (i.e. $r \to 0$):

- $J_m(u)$ remains finite,
- $Y_m(u)$ diverges.

Since the physical displacement of the membrane must remain finite at the centre, we must discard the $Y_m(u)$ solution. Hence:

$$\boxed{R(r) = J_m(kr).} \tag{30}$$

(ii) Special case: $m = 0$

For $m = 0$, the Bessel equation becomes:

$$\frac{d^2R}{du^2} + \frac{1}{u}\frac{dR}{du} + R = 0. \tag{31}$$

Multiplying through by u:

$$u\frac{d^2R}{du^2} + \frac{dR}{du} + uR = 0. \tag{32}$$

This matches the form given in the question and is identical to the equation encountered in the hanging string problem. Therefore, the solution is:

$$\boxed{R(r) = J_0(kr).} \tag{33}$$

(d) Eigenfrequencies of the Circular Membrane

We now apply the boundary condition to determine the allowed frequencies of oscillation.

(i) Quantisation of k and eigenfrequencies

From the previous parts, the general solution for the membrane displacement is:

$$g(r, \phi, t) = J_m(kr)\left[C\cos(m\phi) + D\sin(m\phi)\right]\left[A\cos(\omega t) + B\sin(\omega t)\right], \tag{34}$$

where J_m is the Bessel function of the first kind of order m.

Boundary condition: The membrane is clamped at the rim $r = R$, so:

$$g(R, \phi, t) = 0 \quad \text{for all } \phi, t. \tag{35}$$

Since the angular and time-dependent parts are not identically zero, the condition must be satisfied by the radial part:

$$J_m(kR) = 0. \tag{36}$$

Zeros of Bessel functions: Let α_{mn} denote the n-th positive zero of $J_m(u)$:

$$J_m(\alpha_{mn}) = 0, \quad n = 1, 2, 3, \dots \tag{37}$$

Thus, the allowed values of k are:

$$k_{mn}R = \alpha_{mn} \quad \Rightarrow \quad \boxed{k_{mn} = \frac{\alpha_{mn}}{R}}. \tag{38}$$

Eigenfrequencies: Recall that:

$$k = \frac{\omega}{v}, \quad \text{where } v = \sqrt{\frac{T}{\sigma}}. \tag{39}$$

Hence:

$$\omega_{mn} = vk_{mn} = v\frac{\alpha_{mn}}{R}. \tag{40}$$

$$\boxed{\omega_{mn} = \frac{v\alpha_{mn}}{R}}$$

Physical meaning of m and n

- The integer m determines the number of **nodal diameters**, i.e. straight lines passing through the centre along which the displacement is always zero.
- The integer n determines the number of **nodal circles**, i.e. concentric circles (excluding the boundary) where the displacement vanishes.

Thus, each mode (m, n) corresponds to a distinct vibration pattern of the membrane.

(ii) Frequency ratios and harmonic comparison

We are given the first few zeros of Bessel functions:

n	1	2	3	4
J_0	2.405	5.520	8.654	11.792
J_1	3.832	7.016	10.173	13.324
J_2	5.136	8.417	11.620	14.796

We compute the ratios:

$$\omega_{01} : \omega_{11} : \omega_{21} : \omega_{02}.$$

Since $\omega_{mn} \propto \alpha_{mn}$, the ratios are:

$$\omega_{01} : \omega_{11} : \omega_{21} : \omega_{02} = \alpha_{01} : \alpha_{11} : \alpha_{21} : \alpha_{02}. \tag{41}$$

Substituting values:

$$= 2.405 : 3.832 : 5.136 : 5.520.$$

Normalising by ω_{01}:

$$\frac{\omega_{11}}{\omega_{01}} = \frac{3.832}{2.405} \approx 1.59, \tag{42}$$

$$\frac{\omega_{21}}{\omega_{01}} = \frac{5.136}{2.405} \approx 2.14, \tag{43}$$

$$\frac{\omega_{02}}{\omega_{01}} = \frac{5.520}{2.405} \approx 2.29. \tag{44}$$

Thus:

$$\boxed{\omega_{01} : \omega_{11} : \omega_{21} : \omega_{02} \approx 1 : 1.59 : 2.14 : 2.29.}$$

Comparison with a stretched string

For a stretched string, the harmonic frequencies are:

$$1 : 2 : 3 : 4,$$

i.e. exact integer multiples of the fundamental frequency.

In contrast, the circular membrane has:

$$1 : 1.59 : 2.14 : 2.29,$$

which are **not integer multiples**.

Physical interpretation (why a tabla sounds "thuddy")

Because the overtones of a circular membrane are not harmonically related, the frequencies do not align with a simple harmonic series. As a result:

- The ear cannot combine the overtones into a single definite pitch.
- The sound contains many inharmonic components.
- This produces a characteristic **percussive, "thuddy" sound** rather than a clear musical note.

Thus, the non-harmonic nature of the eigenfrequencies, determined by the zeros of Bessel functions, explains the acoustic behaviour of drums such as the tabla.

3. Symmetry and Simplification

We consider waves on a 2D surface described by the wave equation:

$$\frac{\partial^2 g}{\partial t^2} = v^2 \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial g}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 g}{\partial \phi^2} \right]. \tag{45}$$

(a) A pebble dropped at the origin produces a disturbance that is identical in all directions. This implies **circular symmetry**.

Thus, the displacement g does not depend on the angular coordinate ϕ, i.e.

$$g = g(r, t). \tag{46}$$

Hence:

$$\frac{\partial g}{\partial \phi} = 0, \quad \frac{\partial^2 g}{\partial \phi^2} = 0. \tag{47}$$

Since the system is invariant under rotations about the origin, the displacement must be the same for all values of ϕ.

(ii) Substituting $\frac{\partial^2 g}{\partial \phi^2} = 0$ into the wave equation, we obtain:

$$\frac{\partial^2 g}{\partial t^2} = v^2 \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial g}{\partial r} \right). \tag{48}$$

This is the **radially symmetric wave equation** in two dimensions.

(iii) We look for oscillatory solutions of the form:

$$g(r, t) = A(r) e^{i\omega t}. \tag{49}$$

Therefore,

$$\frac{\partial^2 g}{\partial t^2} = -\omega^2 A(r) e^{i\omega t}. \tag{50}$$

$$\frac{\partial g}{\partial r} = A'(r) e^{i\omega t}, \tag{51}$$

$$r \frac{\partial g}{\partial r} = r A'(r) e^{i\omega t}, \tag{52}$$

$$\frac{\partial}{\partial r} \left(r \frac{\partial g}{\partial r} \right) = \frac{d}{dr} \left(r A'(r) \right) e^{i\omega t}. \tag{53}$$

Thus:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial g}{\partial r} \right) = \frac{1}{r} \frac{d}{dr} \left(r A'(r) \right) e^{i\omega t}. \tag{54}$$

Substituting these values into wave equation:

$$-\omega^2 A(r)e^{i\omega t} = v^2 \left[\frac{1}{r}\frac{d}{dr}\left(rA'(r)\right)e^{i\omega t}\right]. \tag{55}$$

Cancel $e^{i\omega t}$:

$$-\omega^2 A(r) = v^2 \frac{1}{r}\frac{d}{dr}\left(rA'(r)\right). \tag{56}$$

Divide by v^2:

$$\frac{1}{r}\frac{d}{dr}\left(rA'(r)\right) + \frac{\omega^2}{v^2}A(r) = 0. \tag{57}$$

Define:

$$k = \frac{\omega}{v}. \tag{58}$$

Final result:

$$\boxed{\frac{1}{r}\frac{d}{dr}\left(r\frac{dA}{dr}\right) + k^2 A = 0.} \tag{59}$$

This is the radial differential equation governing wave propagation in a two-dimensional radially symmetric system.

(b) From part (a), the radial function $A(r)$ satisfies:

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dA}{dr}\right) + k^2 A = 0. \tag{60}$$

Solving the derivative:

$$\frac{d}{dr}\left(r\frac{dA}{dr}\right) = r\frac{d^2A}{dr^2} + \frac{dA}{dr}. \tag{61}$$

Substituting back into the equation:

$$\frac{1}{r}\left(r\frac{d^2A}{dr^2} + \frac{dA}{dr}\right) + k^2 A = 0. \tag{62}$$

$$\frac{d^2A}{dr^2} + \frac{1}{r}\frac{dA}{dr} + k^2 A = 0. \tag{63}$$

substituting $u = kr$
Let:

$$u = kr. \tag{64}$$

We now transform derivatives with respect to r into derivatives with respect to u.

$$\frac{d}{dr} = \frac{d}{du}\frac{du}{dr} = k\frac{d}{du}. \tag{65}$$

$$\begin{aligned}
\frac{d^2}{dr^2} &= \frac{d}{dr}\left(k\frac{d}{du}\right) && (66)\\
&= k\frac{d}{dr}\left(\frac{d}{du}\right) && (67)\\
&= k\left(\frac{d}{du}\frac{d}{du}\right)\frac{du}{dr} && (68)\\
&= k\left(\frac{d^2}{du^2}\right)k && (69)\\
&= k^2\frac{d^2}{du^2}. && (70)
\end{aligned}$$

Substituting the values into the differential equation,

$$\frac{d^2A}{dr^2} = k^2 \frac{d^2A}{du^2}, \tag{71}$$

$$\frac{1}{r}\frac{dA}{dr} = \frac{1}{r}\left(k\frac{dA}{du}\right). \tag{72}$$

Since $r = \frac{u}{k}$:

$$\frac{1}{r} = \frac{k}{u}. \tag{73}$$

Thus:

$$\frac{1}{r}\frac{dA}{dr} = \frac{k}{u} \cdot k\frac{dA}{du} = \frac{k^2}{u}\frac{dA}{du}. \tag{74}$$

Substituting everything:

$$k^2\frac{d^2A}{du^2} + \frac{k^2}{u}\frac{dA}{du} + k^2 A = 0. \tag{75}$$

Divide through by k^2:

$$\frac{d^2A}{du^2} + \frac{1}{u}\frac{dA}{du} + A = 0. \tag{76}$$

$$\boxed{\frac{d^2A}{du^2} + \frac{1}{u}\frac{dA}{du} + A = 0}$$

This is **Bessel's equation of order zero**.
The general solution is:

$$A(u) = c_1 J_0(u) + c_2 Y_0(u), \tag{77}$$

where $J_0(u)$ and $Y_0(u)$ are Bessel functions of the first and second kind.

Physical condition at $r = 0$

As $u \to 0$ (i.e. $r \to 0$):

- $J_0(u)$ remains finite,
- $Y_0(u)$ diverges logarithmically.

Since the physical displacement must remain finite at the origin, we must discard the $Y_0(u)$ solution. Therefore, the solution becomes:

$$\boxed{A(r) = c_1 J_0(kr).} \tag{78}$$

(c) From part (b), the radial function $A(r)$ satisfies:

$$\frac{d^2A}{dr^2} + \frac{1}{r}\frac{dA}{dr} + k^2 A = 0. \tag{79}$$

We now make the substitution:

$$A(r) = \frac{f(r)}{\sqrt{r}} = f(r)\, r^{-1/2}. \tag{80}$$

$$\frac{dA}{dr} = \frac{d}{dr}\left(f(r)\, r^{-1/2}\right) \tag{81}$$

$$= f'(r)\, r^{-1/2} + f(r)\,\frac{d}{dr}(r^{-1/2}). \tag{82}$$

Now:

$$\frac{d}{dr}(r^{-1/2}) = -\frac{1}{2}r^{-3/2}. \tag{83}$$

Thus:

$$\frac{dA}{dr} = \frac{f'(r)}{\sqrt{r}} - \frac{1}{2}\frac{f(r)}{r^{3/2}}. \tag{84}$$

Differentiating again:

$$\frac{d^2A}{dr^2} = \frac{d}{dr}\left(\frac{f'}{\sqrt{r}} - \frac{1}{2}\frac{f}{r^{3/2}}\right). \tag{85}$$

Differentiating term by term.

$$\frac{d}{dr}\left(\frac{f'}{\sqrt{r}}\right) = f''\,r^{-1/2} + f'\left(-\frac{1}{2}r^{-3/2}\right) \tag{86}$$

$$= \frac{f''}{\sqrt{r}} - \frac{1}{2}\frac{f'}{r^{3/2}}. \tag{87}$$

$$\frac{d}{dr}\left(-\frac{1}{2}\frac{f}{r^{3/2}}\right) = -\frac{1}{2}\left(f'r^{-3/2} + f\cdot\frac{d}{dr}(r^{-3/2})\right). \tag{88}$$

$$\frac{d}{dr}\left(-\frac{1}{2}\frac{f}{r^{3/2}}\right) = -\frac{1}{2}f'r^{-3/2} + \frac{3}{4}fr^{-5/2}. \tag{89}$$

Combining the terms:

$$\frac{d^2A}{dr^2} = \frac{f''}{\sqrt{r}} - \frac{1}{2}\frac{f'}{r^{3/2}} - \frac{1}{2}\frac{f'}{r^{3/2}} + \frac{3}{4}\frac{f}{r^{5/2}} \tag{90}$$

$$= \frac{f''}{\sqrt{r}} - \frac{f'}{r^{3/2}} + \frac{3}{4}\frac{f}{r^{5/2}}. \tag{91}$$

Substituting into the differential equation

$$\left(\frac{f''}{\sqrt{r}} - \frac{f'}{r^{3/2}} + \frac{3}{4}\frac{f}{r^{5/2}}\right) + \frac{1}{r}\left(\frac{f'}{\sqrt{r}} - \frac{1}{2}\frac{f}{r^{3/2}}\right) + k^2\frac{f}{\sqrt{r}} = 0. \tag{92}$$

$$\frac{f''}{\sqrt{r}} - \frac{f'}{r^{3/2}} + \frac{3}{4}\frac{f}{r^{5/2}} + \frac{f'}{r^{3/2}} - \frac{1}{2}\frac{f}{r^{5/2}} + k^2\frac{f}{\sqrt{r}} = 0. \tag{93}$$

$$\frac{f''}{\sqrt{r}} + \left(\frac{3}{4} - \frac{1}{2}\right)\frac{f}{r^{5/2}} + k^2\frac{f}{\sqrt{r}} = 0. \tag{94}$$

$$\frac{f''}{\sqrt{r}} + \frac{1}{4}\frac{f}{r^{5/2}} + k^2\frac{f}{\sqrt{r}} = 0. \tag{95}$$

Multiplying through by $\sqrt{r}$

$$f'' + \frac{1}{4r^2}f + k^2 f = 0. \tag{96}$$

Final Result

$$\boxed{f'' + \left(k^2 - \frac{1}{4r^2}\right)f = 0.}$$

(d)Far-Field Approximation

From part (c), we obtained the differential equation:

$$f'' + \left(k^2 - \frac{1}{4r^2}\right)f = 0. \tag{97}$$

We consider the region far from the source:

$$kr \gg 1 \quad \Rightarrow \quad r \text{ is large.}$$

In this limit:

$$\frac{1}{4r^2} \ll k^2.$$

Hence we neglect the term $\frac{1}{4r^2}$, and the equation simplifies to:

$$f'' + k^2 f = 0. \tag{98}$$

This is the standard simple harmonic equation. Its general solution is:

$$f(r) = C_1 e^{ikr} + C_2 e^{-ikr}. \tag{99}$$

Recall the substitution:

$$A(r) = \frac{f(r)}{\sqrt{r}}. \tag{100}$$

Substitute $f(r)$:

$$A(r) = \frac{C_1 e^{ikr} + C_2 e^{-ikr}}{\sqrt{r}}. \tag{101}$$

The two terms represent:

- e^{ikr}: outgoing wave (moving away from the source),
- e^{-ikr}: incoming wave (moving toward the source).

Since the disturbance originates from the source and propagates outward, we keep only the outgoing wave. Thus:

$$A(r) \approx \frac{C\, e^{ikr}}{\sqrt{r}}. \tag{102}$$

Recall:

$$g(r,t) = A(r) e^{i\omega t}.$$

Substitute $A(r)$:

$$g(r,t) \approx \frac{C}{\sqrt{r}} e^{i(kr - \omega t)}. \tag{103}$$

Final Result

$$\boxed{g(r,t) \approx \frac{C}{\sqrt{r}} e^{i(kr-\omega t)}}$$

This shows that the wave propagates outward as $e^{i(kr-\omega t)}$,
(e) From part (d), the far-field solution for the wave is:

$$g(r,t) \approx \frac{C}{\sqrt{r}} e^{i(kr - \omega t)}. \tag{104}$$

The amplitude of the wave is given by the magnitude of the prefactor:

$$A(r) \propto \frac{1}{\sqrt{r}}. \tag{105}$$

Thus, in two dimensions:

$$\boxed{A(r) \propto \frac{1}{\sqrt{r}}.}$$

We now compare how amplitude depends on distance in different spatial dimensions.

Dimension	**Wavefront Geometry**	**Amplitude**
1D (string)	flat (plane)	constant
2D (pond surface)	circle	$\frac{1}{\sqrt{r}}$
3D (sound, light)	sphere	$\frac{1}{r}$

- In **1D**, the wave does not spread sideways, so the amplitude remains constant.
- In **2D**, the wave spreads along a growing circle, causing the amplitude to decrease as $1/\sqrt{r}$.
- In **3D**, the wave spreads over a spherical surface, leading to a faster decay of amplitude as $1/r$.

Thus,

$$\boxed{\text{In 2D, the amplitude of the wave decays as } A(r) \propto \frac{1}{\sqrt{r}}.}$$

(f) We now confirm the amplitude dependence using energy conservation.

As the wave propagates outward, the total energy carried by the wave remains constant. This energy is distributed over the entire wavefront.

The **intensity** I (energy per unit length/area per unit time) is proportional to the square of the amplitude:

$$I \propto A^2. \tag{106}$$

Since the total power crossing any wavefront is conserved, we have:

$$I \times (\text{size of wavefront}) = \text{constant}. \tag{107}$$

In 2D, the wave spreads out as a circle of radius r.
The length of the wavefront is:

$$L = 2\pi r \propto r. \tag{108}$$

Thus:

$$I \times r = \text{constant}. \tag{109}$$

Substituting $I \propto A^2$:

$$A^2 \cdot r = \text{constant}. \tag{110}$$

Hence:

$$A^2 \propto \frac{1}{r} \quad \Rightarrow \quad \boxed{A \propto \frac{1}{\sqrt{r}}.} \tag{111}$$

(i) One dimension (1D) The wave does not spread; the wavefront is effectively a point (or constant cross-section).

Thus:

$$I = \text{constant} \quad \Rightarrow \quad A = \text{constant}. \tag{112}$$

(ii) Three dimensions (3D) In 3D, the wave spreads over a sphere.

The surface area of the wavefront is:

$$S = 4\pi r^2 \propto r^2. \tag{113}$$

Thus:

$$I \cdot r^2 = \text{constant}. \tag{114}$$

So:

$$A^2 \propto \frac{1}{r^2} \quad \Rightarrow \quad \boxed{A \propto \frac{1}{r}.} \tag{115}$$

Dimension	**Wavefront size**	**Amplitude**
1D	constant	constant
2D	$\propto r$	$\frac{1}{\sqrt{r}}$
3D	$\propto r^2$	$\frac{1}{r}$

$$\boxed{A \propto \begin{cases} \text{constant}, & \text{1D} \\ \frac{1}{\sqrt{r}}, & \text{2D} \\ \frac{1}{r}, & \text{3D} \end{cases}}$$

The amplitude decay derived from energy conservation agrees exactly with the result obtained from solving the wave equation in part (d), confirming the consistency of the analysis.
